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Do tax cuts generate twin deficits? A multi-country analysis

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Abstract. We study the effects of tax shocks on the budget and external deficits for 16 industrialized countries over the post-1970 period. Our structural approach is based on a small open economy model where a tax cut affects the external deficit by two distinct channels. The demographic channel works through the overlapping-generation structure of the model. The forecasting channel works through the dynamic structure of the model. Our empirical analysis documents that tax shocks generate twin deficits, and that both channels play important roles in explaining the positive comovement between the budget and external deficits. JEL classification: E62, F32, F41

Est-ce que des réductions d'impôt engendrent des déficits jumeaux? Une analyse de plusieurs pays. On étudie des effets de chocs fiscaux sur les déficits budgétaire et extérieur de 16 pays industrialisés au cours de la période de l'après 1970. L'approche structurelle adoptée est basée sur un modèle de petite économie ouverte où les effets d'une réduction d'impôt affectent le déficit extérieur par deux canaux distinct : le canal démographique joue à travers la structure de chevauchement des générations, et le canal prévisionnel joue à travers la structure dynamique du modèle. On montre empiriquement que les chocs fiscaux engendrent les déficits jumeaux, et que les deux canaux jouent un rôle important dans l'explication du co-mouvement positif entre les déficits budgétaire et extérieur.

1. Introduction

The last few years have seen renewed interest in the impact of government budget deficits on the economy, and in particular on external deficits (e.g., Bartolini and Lahiri 2006; Beetsma, Klaassen, and Giuliadori 2008; Bussiere, Fratzcher, and

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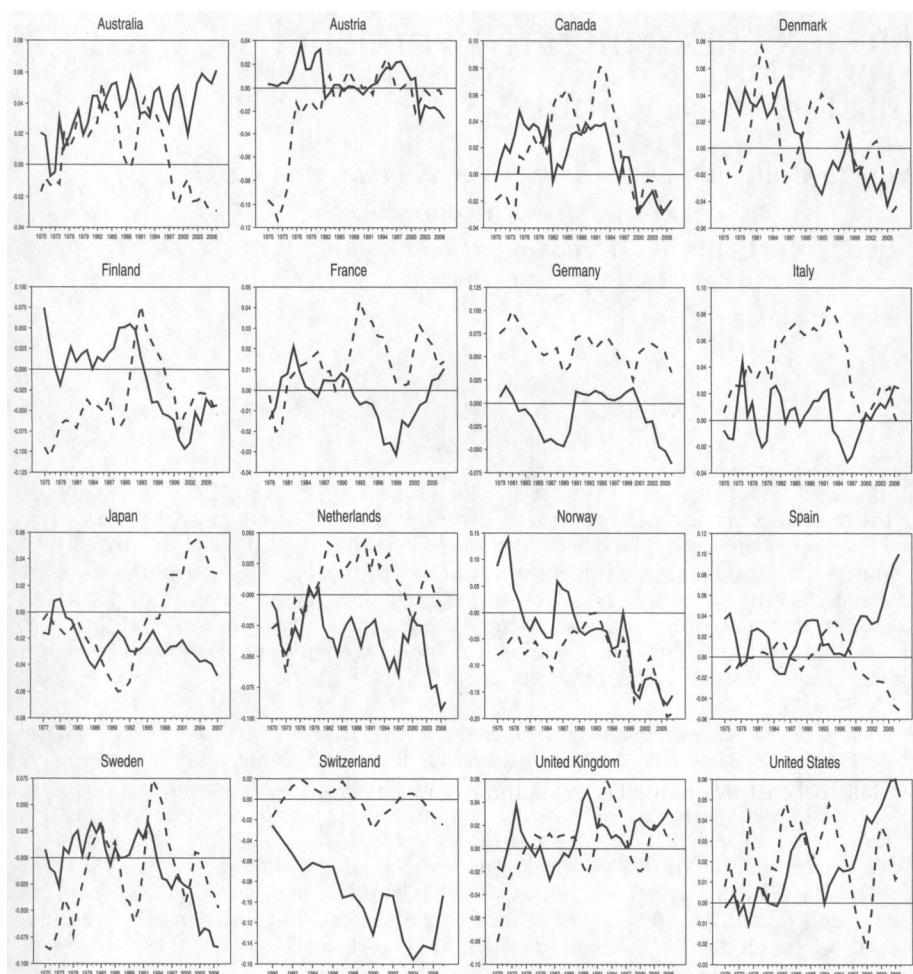


FIGURE 1 Budget and external deficit ratios

NOTE: The solid lines show the external deficit to output ratios, while the dashed lines show the budget deficit to output ratios.

Muller 2010; Chinn 2005; Corsetti and Muller 2006 and 2008; Kim and Roubini 2008; Monacelli and Perotti 2010). One view is that large budget deficits reduce aggregate saving and generate external deficits. The empirical support for this view in international data, however, is slim. As shown in figure 1, the budget and external deficits series do not display a homogeneous pattern of positive comovements among industrialized countries. In part, the lack of homogeneous evidence occurs because the raw data jointly capture the effects of several shocks. These shocks may simultaneously push the budget and external deficits in different directions.

To circumvent this difficulty, our analysis relies on a structural approach to isolate solely the effects of tax shocks on the budget and external deficits. This exercise is in the spirit of the classical literature seeking to discriminate between the twin deficit hypothesis and the Ricardian equivalence hypothesis. The twin deficit hypothesis stipulates that a tax cut raises both the budget and external deficits. In contrast, the Ricardian equivalence hypothesis stipulates that a tax cut raises the budget deficit, but leaves the external deficit unchanged. Also, our focus on tax shocks distinguishes our analysis from several recent papers that focus on government spending or budget deficit shocks (e.g., Beetsma, Klaassen, and Giuliodori 2008; Corsetti and Muller 2008; Kim and Roubini 2008; and Monacelli and Perotti 2010).

Our structural approach employs a tractable small open economy model with overlapping generations. The model is tractable in the sense that the analytical solution of the equilibrium yields a vector error correction representation that includes both the budget and external deficits, as well as other variables. For our purpose, the model nests both the twin deficit and Ricardian equivalence hypotheses. More important, it also nests two distinct channels by which the budget deficit may affect the external deficit. In particular, an increase in the external deficit can happen via a demographic channel and via a forecasting channel. The demographic channel is standard in overlapping-generation models. In these models, consumers can shift a portion of their future tax burden to unborn generations. As a result, a tax cut raises consumer's wealth and aggregate consumption, lowers aggregate saving, and raises the external deficit.

The forecasting channel is less standard, but it occurs in most dynamic models. In such models, agents require forecasts of future variables to make optimal choices, and rational agents use all the relevant available information to make these forecasts. Then, as long as the budget deficit contains information useful to forecast future variables, it may affect optimal choices and thus the external deficit.

To motivate the forecasting channel further, consider the following examples. First, it is well established that some elements related to the budget (e.g., government expenditures, military spending, and political parties in power) Granger-cause Solow residuals (Evans 1992; Hall 1988). To be concrete, consider the example where an increase in the budget deficit due to an expansion of military spending eventually improves technology. This expansion would signal a future increase in output. As is standard, higher future output raises consumer's wealth. This in turn raises consumption, lowers aggregate saving, and raises the external deficit. Thus, this forecasting channel suggests that an increase in the budget deficit raises the external deficit. We can also consider the example where some tax reductions that raise the budget deficit improve entrepreneurship and innovations. The resulting higher innovation would signal higher future output. Again, the higher future output would raise consumption, lower aggregate saving, and raise the external deficit.

Second, it is also fairly well documented that the public debt-to-output ratio is mean reverting (Bohn 1998). This suggests that there are mechanisms by which an increase in the current budget deficit signals a future reduction in the budget deficit. To be concrete, consider the example where a current tax reduction starves the beasts and eventually forces future reductions in government expenditures (Bartlett 2007; Romer and Romer 2009). If this starve-the-beast hypothesis holds, a tax cut signals lower future government expenditures. As is standard, the lower future government expenditures raise consumer's wealth. In turn, the higher wealth raises aggregate consumption, lowers aggregate saving, and raises the external deficit.

The insight of these examples is that if agents can improve their forecasts of future output and government expenditures by using an information set that includes elements related to the current budget deficit, then the budget deficit will affect the external deficit. The previous discussion should also make clear that the forecasting channel does not necessarily imply a positive relation between the budget and external deficits. For example, a current increase in the budget deficit may reflect fiscal illusion (Buchanan and Wagner 1977; Niskanen 1978) or fiscal irresponsibility (Gale and Orzag 2004). In this context, an increase in the budget deficit would signal higher future government expenditures. This, of course, would lower consumer's wealth, lower consumption, raise saving, and lower the external deficit.

Our analysis improves on the existing literature in several ways. First, because of its tractability, our approach permits a comprehensive study. Our empirical work uses data for 16 industrialized countries over the post-1970 period. The sample includes Australia, Austria, Canada, Denmark, Finland, France, Germany, Italy, Japan, Netherlands, Norway, Spain, Sweden, Switzerland, the United Kingdom, and the United States.

Second, our approach allows us to isolate tax shocks. The ability to eliminate the effects of other shocks is crucial to verify whether tax shocks generate positive comovements between the budget and external deficits. To highlight the importance of this exercise, we report the correlation between the deficits for several cases. For the first case, we simply compute the correlation from the raw data. For our countries, these unconditional correlations do not display a specific pattern. For example, we document a large positive and statistically significant unconditional correlation between the budget and external deficits for Canada and Norway, a small and insignificant unconditional correlation for Australia and the United States, and a large negative and significant unconditional correlation for Spain.

For the second case, we compute correlations using vector error correction specifications that are unrestricted, but hit only by tax shocks. The empirical specification is unrestricted in the sense that it does not embed all the restrictions of the structural model. This specification is useful to construct the required data series that would have prevailed if only tax shocks occur. The tax shocks are identified by a Cholesky decomposition, because this decomposition fully conforms

with the structural model. In particular, this identification strategy purges the effects of the business cycle on taxes and transfers to retain only the exogenous portion of unexpected net tax revenues. Empirically, the resulting conditional correlations display an obvious pattern consistent with twin deficits. The estimates are usually large, positive, and significant. This homogeneity is robust to cases that ignore the error correction term and to cases that use an alternative strategy to identify tax shocks. Interestingly, this homogeneity contrasts with the results reported in the government spending shocks literature. For example, Kim and Roubini (2008) find a negative correlation using an identification strategy based on a Cholesky decomposition, while Monacelli and Perotti (2010) find a positive correlation using an identification strategy based on Blanchard and Perotti (2002).

For the third case, we compute the conditional correlations using restricted vector error correction specifications that embed all the restrictions of the structural model and are hit only by tax shocks. Again, the resulting conditional correlations display an obvious pattern consistent with twin deficits. The conditional correlations are large, positive, and significant for all countries. The analysis reveals that the large conditional correlations are attributable to the positive responses of both deficits to a tax cut over most horizons. A tax cut generates a positive impact response of the budget deficit for all countries, and a positive impact response of the external deficit for most countries. For the countries that do not display the positive impact response, the external deficit rises sharply soon after impact. Over time, the positive responses of the budget and external deficits are persistent. In addition, we document that the tax cut has important economic effects on the public and external debts for all countries, and that it accounts for a substantial portion of the unexpected fluctuations of the budget and external deficits.

Finally, the main distinguishing feature of our paper is that the structural approach permits the evaluation of the different channels by which tax shocks affect the external deficit. From a theoretical perspective, the demographic channel implies positive comovements between the budget and external deficits. The forecasting channel, however, may produce either positive or negative comovements. Empirically, our analysis reveals that both demographic and forecasting channels are important to understand the mechanisms by which a tax cut generates positive comovements between the budget and external deficits. Both channels work to produce positive conditional correlations between the deficits, and to propagate the tax shocks through time. We document that the forecasting channel explains the shape of the responses of the external deficit following a tax cut, while the demographic channel explains the amplitude of the responses.

The paper is organized as follows. Section 2 presents the structural model and discusses the different channels by which a tax cut affects the external deficit. Section 3 presents our empirical approach. Section 4 discusses our empirical results. Section 5 concludes.

2. A small open economy

Our small open economy model is based on the overlapping-generation model of Blanchard (1985). The model yields solutions where the budget deficit affects the external deficit via two distinct channels.

2.1. The model

The demographic structure of the small open economy is as follows. The birth rate is denoted by p and the survival probability by $(1 - p)$. At period t , the size of a cohort that was born at time $s \leq t$ is $P_{s,t} = p(1 - p)^{t-s}$, where $P_{s,s} = p$ and total population is $P = \sum_{s=-\infty}^t P_{s,t} = 1$. As a result, the economy is populated by overlapping generations of consumers when $0 < p < 1$, but populated by a representative infinitely lived consumer when $p = 0$. Also, note that, as a convention, variables indexed by s and t refer to cohort-specific variables, while variables indexed only by t refer to aggregate variables.

At time t , a consumer from the cohort who was born at time s chooses a stream of consumption to solve

$$\max E_t \left[\sum_{j=0}^{\infty} \left(\frac{1}{1 + \rho} \right)^j U(C_{s,t+j}) \right], \quad (1)$$

subject to

$$A_{s,t+1} = (1 + \rho)(A_{s,t} + L_{s,t} - T_{s,t} - C_{s,t}). \quad (2)$$

The term E_t is the expectation operator conditional on period t information; the function $U(\cdot)$ denotes the period utility; $C_{s,t}$ is consumption; $A_{s,t} = (B_{s,t} + F_{s,t})$ is (non-human) wealth, where $B_{s,t}$ and $F_{s,t}$ are the quantities purchased of one-period home government and foreign bonds; $T_{s,t}$ is lump-sum taxes; and $L_{s,t}$ is a non-insurable stochastic (labour plus dividend) income. Note that each consumer inelastically supplies a unit of labour, such that labour supply equals total population. As is standard, we assume the existence of insurance firms that make annuity payments to the consumer who holds wealth during his life. In exchange, the insurance firms inherit the wealth at the consumer's death. This implies that the gross return on wealth is $(1 + \rho) = (1 + r)/(1 - p)$, where $(1 + r)$ is the world real gross return on one-period bonds.

At time t , the firm chooses employment and investment to solve

$$\max E_t \left[\sum_{j=0}^{\infty} \left(\frac{1}{1 + r} \right)^j \mathcal{D}_{t+j} \right], \quad (3)$$

subject to

$$\mathcal{D}_t = Y_t - w_t N_t - I_t \quad (4)$$

$$Y_t = Z_t K_t^\alpha N_t^{1-\alpha} \quad (5)$$

$$K_{t+1} = I_t + (1 - \delta)K_t, \quad (6)$$

where $\mathcal{D}_t$ is dividends, Y_t is output, w_t is the wage rate, N_t is employment, I_t is investment, Z_t is a stochastic productivity shock, and K_t is the capital stock. Aggregate income is defined as $L_t \equiv w_t N_t + \mathcal{D}_t = Y_t - I_t$. In the competitive equilibrium, $N_t = P = 1$.

The home government faces the budget constraint

$$B_{t+1} + B_{t+1}^* = (1 + r)(B_t + B_t^* + G_t - T_t), \quad (7)$$

where B_t^* is the quantity purchased of one-period home government bonds by foreign consumers, and G_t is the home government stochastic expenditures on goods and services. The budget deficit is defined as

$$D_t = \left(\frac{r}{1+r} \right) (B_t + B_t^*) + G_t - T_t. \quad (8)$$

The term $[r/(1+r)](B_t + B_t^*)$ denotes payments to service the debt and $(G_t - T_t)$ is the primary deficit.

The external deficit is defined as the negative of the current account. The current account is measured as

$$CA_t = \left(\frac{r}{1+r} \right) (F_t - B_t^*) + Q_t - C_t. \quad (9)$$

The term $[r/(1+r)](F_t - B_t^*)$ denotes net income from foreign assets and $(Q_t - C_t)$ is the trade balance, where $Q_t \equiv Y_t - I_t - G_t$ is net output (output net of investment and government expenditures).

2.2. The model's solution

We solve the model as follows (see the technical appendix, A.1 and A.2, available on the CJE website). First, assuming quadratic preferences, we solve the consumer's problem to obtain cohort-specific consumption functions. Second, we aggregate across consumers to get the aggregate consumption function. Third, we use the aggregate consumption function to substitute out consumption from the current account equation (9). This yields the optimal rule for the current account:

$$CA_t = - \left(\frac{p}{1+r} \right) A_t - \left(\frac{\rho - r}{\rho} \right) D_t + x_t, \quad (10)$$

where

$$x_t = - \sum_{j=1}^{\infty} \left(\frac{1}{1+\rho} \right)^j E_t \left[\Delta Q_{t+j} + \left(\frac{\rho-r}{\rho} \right) \Delta D_{t+j} \right]. \quad (11)$$

Equation (10) relates the current account to wealth, the budget deficit, and the adjusted current account x_t . Equation (11) states that the adjusted current account represents the purely forward-looking component of the optimal rule for the current account. Note that the forward-looking component depends exclusively on two forcing variables: the change in net output $\Delta Q_t \equiv Q_t - Q_{t-1}$ and the change in the budget deficit $\Delta D_t \equiv D_t - D_{t-1}$. Note that the current change in net output is unaffected by tax policies (see the technical appendix, A.2).

Fourth, the optimal rule for aggregate wealth is derived from the aggregate consumption function and the aggregate budget constraint. The result is

$$A_{t+1} = (1-p)A_t + (1+r) \left(\frac{r}{\rho} \right) D_t + w_{t+1}, \quad (12)$$

where

$$w_{t+1} = -(1+r) \sum_{j=1}^{\infty} \left(\frac{1}{1+\rho} \right)^j E_t \left[\Delta Q_{t+j} + \left(\frac{\rho-r}{\rho} \right) \Delta D_{t+j} \right] \quad (13)$$

and w_{t+1} denotes the adjusted wealth.

For completeness, the optimal rules for the current account and aggregate wealth are coupled with a law of motion for forcing variables. For now, we model the law of motion that drives these variables as a first-order vector autoregression (VAR):

$$\begin{pmatrix} \Delta Q_{t+1} \\ \Delta D_{t+1} \end{pmatrix} = \begin{pmatrix} \pi_{11} & \pi_{12} \\ \pi_{21} & \pi_{22} \end{pmatrix} \begin{pmatrix} \Delta Q_t \\ \Delta D_t \end{pmatrix} + \begin{pmatrix} v_{Q,t+1} \\ v_{D,t+1} \end{pmatrix}$$

or

$$H_{t+1} = \Pi H_t + V_{t+1}. \quad (14)$$

Note that, as is standard, we posit the forcing variables in terms of net output and the budget deficit, rather than in terms of the underlying exogenous shocks. We do so because net output and the budget deficit are easily observable, and because both are functions of only exogenous variables in equilibrium. This approach is widely used, and it has the advantage of not requiring an explicit specification of the exogenous stochastic process that generates the unobservable shocks (see, e.g., section 2.3 of Obstfeld and Rogoff 1996). In our context, the approach is valid for the following reasons. First, in the small open economy model

with inelastic labour, equilibrium output, Y_t , and investment, I_t , only depend contemporaneously on the exogenous interest rate, r , and the exogenous stochastic productivity shock, Z_t (see the technical appendix, A.2). Then, allowing for exogenous stochastic government expenditures, G_t , equilibrium net output, Q_t , is a function of only exogenous variables. Second, the current budget deficit, D_t , is a function of only the exogenous government expenditures, G_t , and lump-sum taxes, T_t , as well as the interest rate, r .

We use the process (14) to construct the forecasts of future forcing variables in the forward-looking components (11) and (13). The adjusted current account reduces to

$$x_t = \theta' H_t \quad (15)$$

or $x_t = \theta_1 \Delta Q_t + \theta_2 \Delta D_t$, where $\theta' = (\theta_1 \ \theta_2)$ and $\theta' = -(e'_1 + [(\rho - r)/\rho]e'_2) [\Pi/(1 + \rho)] [I - \Pi/(1 + \rho)]^{-1}$ for $e_1 = (1 \ 0)'$, $e_2 = (0 \ 1)'$, and I is the identity matrix. Also, adjusted wealth reduces to

$$w_{t+1} = (1 + r)\theta' H_t. \quad (16)$$

2.3. Tax cuts and twin deficits

For a given contemporaneous government expenditure, a tax cut innovation clearly results in a budget deficit. To engineer a twin deficit, the budget deficit must generate an external deficit. The relation between the budget and external deficits are summarized by equations (10) and (11), as well as the solution (15). These expressions embed two different channels by which the budget deficit affects the external deficit. The demographic channel operates via the demographic structure of the model and is controlled by the birth rate p . The forecasting channel operates via the dynamic structure of the law of motion, and is controlled by the parameter π_{12} .

As a benchmark, consider a case where both channels are closed ($p = 0$ and $\pi_{12} = 0$). In this case, $\rho = r$ and equations (10) and (11) reduce to $CA_t = x_t$ and $x_t = -\sum_{j=1}^{\infty} (1 + r)^{-j} E_t[\Delta Q_{t+j}]$. Clearly, the budget deficit does not appear in these equations. In addition, the solution (15) becomes $x_t = \theta_1 \Delta Q_t$, given that $\theta_2 = 0$. Again, the budget deficit does not appear in the solution. As a result, there is no relation between the budget and external deficits. From this benchmark, we can sequentially study the effect of opening only one channel.

The effects of the demographic channel depend on the birth rate p . When $0 < p < 1$, the economy is populated by overlapping generations, and consumers can shift a fraction ($0 < [(\rho - r)/\rho] < 1$) of their tax burden to unborn generations. This ensures that changes in the budget deficit affect the current account. To see this, note that keeping $\pi_{12} = 0$ implies that equations (10) and (11) remain $CA_t = -[p/(1 + r)]A_t - [(\rho - r)/\rho]D_t + x_t$ and $x_t = -\sum_{j=1}^{\infty} (1 + \rho)^{-j} E_t[\Delta Q_{t+j} + ((\rho - r)/\rho)\Delta D_{t+j}]$. Clearly, the budget deficit appears in these equations: the

TABLE 1
Simulated correlations

		<i>p</i>				
		0.00	0.01	0.03	0.05	0.10
Panel A: Demographic channel ($\pi_{12} = 0$)						
π_{22}	−0.90	0.00	0.91	0.90	0.90	0.90
	0.00	0.00	0.99	0.98	0.98	0.98
	0.90	0.00	0.93	0.94	0.94	0.95
		π_{12}				
		−0.70	−0.30	0.00	0.30	0.70
Panel B: Forecasting channel ($p = 0$)						
$\pi_{11} = \pi_{22}$	−0.25	−0.19	−0.18	0.00	0.14	0.15
	0.00	−0.18	−0.21	0.00	0.21	0.80
	0.25	−0.08	−0.21	0.00	0.50	0.99

NOTES: Panel A: $\pi_{11} = \pi_{21} = 0$. Panel B: $\pi_{21} = 0$. Entries are the simulated correlations between the budget deficit ratio and the external deficit ratio. The correlations are constructed using only innovations to the budget deficit.

current level of the budget deficit directly affects the current account and future changes in the budget deficit directly affect the adjusted current account. The latter effect is summarized by the solution (15), where $x_t = \theta_1 \Delta Q_t + \theta_2 \Delta D_t$ with $\theta_2 = -[(\rho - r)/\rho]\pi_{22}/[1 + \rho - \pi_{22}]$. The effects on both the current account and the adjusted current account depend on the fraction, $[(\rho - r)/\rho]$, of the tax burden shifted to future generations.

The demographic channel promotes positive comovements between the budget and external deficits. To see this, note that the net effect is determined by the relative sizes of the effect of the budget deficit on the current account ($-[(\rho - r)/\rho]$) and the effect of changes in the budget deficit on the adjusted current account ($\theta_2 = -[(\rho - r)/\rho]\pi_{22}/[1 + \rho - \pi_{22}]$). It can be shown that $\theta_2 < [(\rho - r)/\rho]$, regardless of the value of π_{22} . Accordingly, the net effect of an increase in the budget deficit is to reduce the current account at impact. This arises because an increase in the budget deficit leads to a positive wealth effect, since consumers eschew a portion of their current tax burden. This wealth effect forces an impact increase in consumption and an impact deterioration of the current account. This impact response suggests a positive correlation between the budget and external deficits.

For illustrative purposes, we verify this intuition from some simulation exercises that account for the dynamics of the budget and external deficits relative to net output (similar transformations will be used in our empirical application). Table 1 shows that the correlation between the budget deficit ratio and the external deficit ratio is positive, regardless of the value of π_{22} , as long as $\rho > 0$. For these computations, we construct simulated series for the current account

from equations (10) and (15), and for the budget deficit and net output from the process (14). To do so, we parameterize Π to various values and focus on the innovation $v_{D,t}$ because it directly affects the budget deficit. Formally, we fix $v_{Q,t} = 0$, and assume that $v_{D,t}$ is drawn from a standard normal distribution.

Next, the effects of the forecasting channel depend on the parameter π_{12} . When $\pi_{12} \neq 0$, changes in the budget deficit Granger-cause changes in net output. As discussed in the introduction, there exists several mechanisms by which changes in the budget deficit contain information useful to forecast future changes in net output. In turn, these mechanisms ensure that changes in the budget deficit affect the current account. To see this, note that keeping $p = 0$ ($\rho = r$) implies that equations (10) and (11) reduce to $CA_t = x_t$ and $x_t = -\sum_{j=1}^{\infty} (1+r)^{-j} E_t[\Delta Q_{t+j}]$. Clearly, the budget deficit does not appear in these equations, so that changes in the budget deficit do not directly affect the current account. However, changes in the budget deficit indirectly affect the current account. That is, equation (15) becomes $x_t = \theta_1 \Delta Q_t + \theta_2 \Delta D_t$, where $\theta_2 = -\pi_{12}/[(1+r-\pi_{11})(1+r-\pi_{22}) - \pi_{12}\pi_{21}]$. Hence, changes in the budget deficit appear in the solution, because changes in the budget deficit indirectly affect the current account via their effect on expected future net output. The indirect effect occurs because changes in the budget deficit contain information useful to forecast future changes in net output.

Importantly, the forecasting channel can promote either positive or negative comovements between the budget and external deficits. This occurs because the direction of the indirect effect is determined by the sign of θ_2 , which is related to the sign of π_{12} . For example, $\theta_2 < 0$ when $\pi_{12} > 0$ and $\pi_{11} = \pi_{22} = \pi_{21} = 0$. In this case, a rise in the budget deficit signals future increases in net output that generate a positive wealth effect, an increase in consumption, and a deterioration of the current account. This mechanism suggests a positive correlation between the budget and external deficits. In contrast, $\theta_2 > 0$ when $\pi_{12} < 0$ and $\pi_{11} = \pi_{22} = \pi_{21} = 0$. In this case, a rise in the budget deficit signals future reductions in net output that generate a negative wealth effect, a decrease in consumption, and an improvement of the current account. This mechanism suggests a negative correlation between the budget and external deficits.

Table 1 confirms this intuition. Namely, the correlation between the budget deficit ratio and the external deficit ratio is negative for negative values of π_{12} . Conversely, the correlation is positive for positive values of π_{12} .

3. The empirical approach

We wish to study the effects of a tax cut on the budget and external deficits of several countries. To do so, we first generalize the model's solution for the current account and aggregate wealth. In particular, we wish to employ a richer law of motion for forcing variables. Using this improved law of motion, we show that the solution can be represented as vector error correction models (VECMs).

Then, we test the restrictions that the structural model imposes on the VECMs. Finally, we discuss how the VECMs can be used to evaluate the relation between the budget and external deficits generated by tax shocks.

3.1. *A VECM representation*

The model's solution represented by equations (15) for the current account and (16) for aggregate wealth result from a combination of the forward-looking decision rules (11) and (13) with the law of motion (14). This solution of course, embeds both the demographic and the forecasting channels. It is worth stressing that the forecasting channel operates via the forecasts of future forcing variables. In this context, it is crucial to specify adequately the law of motion. For this purpose, we improve the law of motion (14) along two important dimensions.

First, we account for the long-run relation between the forcing variables. For this, recall that the law of motion (14) includes the budget deficit and net output in first differences, so that the forecasting channel is operative whenever changes in the budget deficit contain information useful to predict future changes in net output. The formulation in first differences is appropriate because pretesting over post-1970 annual data for the 16 industrialized countries (see the data appendix) confirms that the null hypothesis of unit roots cannot be rejected for the levels of the budget deficit and net output, against the alternative of linear and quadratic deterministic trend stationary processes. However, we improve the law of motion (14) by admitting the possibility that there exists a stable long-run relation between the budget deficit and net output. This possibility accords with the structural model, which imposes that all agents, including the government, are solvent. This possibility also accords with the available empirical evidence. For example, Bohn (1998) argues that the United States primary surplus is an increasing function of the debt-to-output ratio, and that the debt-to-output ratio is mean reverting. This suggests that the budget deficit and net output are cointegrated. To implement this notion, we add an error correction term to the law of motion (14). In turn, this implies a richer version of the forecasting channel, because it introduces another mechanism by which the current budget deficit predicts future net output.

Second, we include an additional variable that summarizes the agents's relevant information set. Recall that the law of motion (14) includes only the forcing variables. We improve the law of motion by admitting the possibility that there exists other exogenous variables that help forecasts changes in net output. In similar circumstances, Campbell and Deaton (1989) and Boileau and Normandin (2002) show that including the purely forward-looking component of the optimal rule allows the econometrician to recover entirely the agents's information set. The insight is that, under the null hypothesis that the model holds, such a component reveals agents's expectations of forcing variables. In our context, we augment the law of motion (14) by including the adjusted current account, given that it is an excellent surrogate for missing exogenous variables.

With these added elements, the law of motion is a VECM:

$$\mathbf{X}_{t+1} = \Gamma \mathbf{X}_t + \beta \mathbf{y}_t + \mathbf{U}_{x,t+1}, \quad (17)$$

where $\mathbf{X}_t = (\Delta Q_t \ \Delta D_t \ x_t)'$, $\mathbf{y}_t = (Q_t \ D_t)'$, and $\mathbf{U}_{x,t} = (u_{Q,t} \ u_{D,t} \ u_{x,t})'$. As a law of motion, (17) includes (14) but adds two terms. It adds the error correction term $\beta \mathbf{y}_t$. It also adds feedbacks between the adjusted current account x_t and the forcing variables.

To proceed, we estimate an unrestricted version of the law of motion (17). To do so, we measure the actual adjusted current account x_t obtained by evaluating (10) using data on the current account, aggregate wealth, and the budget deficit, as well as values for the interest rate and the birth rate. The values of these parameters are discussed in section 3.2. We note that pretesting over all the countries suggests that the first-order specification (17) is a good representation of the data, as suggested by the Aikake and Schwarz criteria and by the Ljung-Box test on the autocorrelation of the residuals.

Then, we use the estimated law of motion (17) to construct the necessary forecasts that appear in the forward-looking solution (11). The reduced form is $\hat{x}_t = v' \mathcal{X}_t$ where $\hat{x}_t$ is the predicted value of the adjusted current account and $\mathcal{X}_t = (Q_t \ D_t \ x_t \ Q_{t-1} \ D_{t-1})'$ (see the technical appendix, A.3). The vector of parameters v is a complex combinations of the estimated parameters of the law of motion (17), as well as ρ and r . The reduced form is used to obtain a restricted version of the VECM:

$$\hat{\mathbf{X}}_{t+1} = \Gamma_x \hat{\mathbf{X}}_t + \beta_x \mathbf{y}_t + \hat{\mathbf{U}}_{x,t+1}, \quad (18)$$

where $\hat{\mathbf{X}}_t = (\Delta Q_t \ \Delta D_t \ \hat{x}_t)'$ and $\hat{\mathbf{U}}_{x,t+1} = (e_1 \ e_2 \ v)' \mathbf{U}_{x,t+1}$, for $e_1 = (1 \ 0 \ 0)'$ and $e_2 = (0 \ 1 \ 0)'$. The matrices Γ_x and β_x are versions of Γ and β that include all the restrictions of the structural model displayed in the vector v .

We use a similar approach to obtain the unrestricted and restricted VECMs for adjusted wealth. First, we posit the law of motion as

$$\mathbf{W}_{t+1} = \Psi \mathbf{W}_t + \eta \mathbf{y}_t + \mathbf{U}_{w,t+1}, \quad (19)$$

where $\mathbf{W}_t = (\Delta Q_t \ \Delta D_t \ w_t)'$. We then use an estimated version of the law of motion to construct the forecasts in the forward-looking solution (13). The result is the reduced-form solution for adjusted wealth: $\hat{w}_t = v'_w \mathcal{W}_t$, where $\mathcal{W}_t = (Q_t \ D_t \ w_t \ Q_{t-1} \ D_{t-1})'$ and $v_w = (1 + r)v$. This solution is then rewritten as a restricted VECM:

$$\hat{\mathbf{W}}_{t+1} = \Psi_w \hat{\mathbf{W}}_t + \eta_w \mathbf{y}_t + \hat{\mathbf{U}}_{w,t+1}, \quad (20)$$

where $\hat{\mathbf{W}}_t = (\Delta Q_t \ \Delta D_t \ \hat{w}_t)'$ and $\hat{\mathbf{U}}_{w,t+1} = (e_1 \ e_2 \ v_w)' \mathbf{U}_{w,t+1}$. The matrices Ψ_w and η_w are versions of Ψ and η that include all the restrictions of the structural model displayed in the vector v_w .

3.2. Birth rates and model restrictions

The restricted reduced forms (18) and (20) are versions of the unrestricted laws of motion (17) and (19) that embed the restrictions of the structural model. The restrictions involve the parameters r and p . In what follows, we calibrate the value of the real interest rate to $r = 0.04$ per year, and we estimate the value of the birth rate p that is consistent with these restrictions. To do so, we exploit the orthogonality restrictions implied by the null hypothesis that the model is valid. Under the null, the predicted and actual adjusted current accounts are the same: $\hat{x}_t = x_t$. This implies that the variable $\varepsilon_t = x_t - (1 + \rho)x_{t-1} - \Delta Q_t - ((\rho - r)/\rho)\Delta D_t$ must be orthogonal to all lagged variables (see the technical appendix, A.4).

Table 2 presents the actual annual birth rate $\wp$ as well as different estimates based on the orthogonality restrictions. The estimates $\underline{p}$ and $\bar{p}$ are the smallest and largest values for which the orthogonality conditions are not rejected at the 1%, 5%, and 10% levels of significance. These bounds are obtained by performing a grid search for the values of the birth rate and selecting those for which all the coefficients associated with the regression of ε_t on x_{t-1} , ΔQ_{t-1} , and ΔD_{t-1} are jointly insignificant. The estimate $\hat{p}$ is a generalized method of moments (GMM) estimate. For this, we exploit the orthogonality between ε_t and a vector that includes a constant, ΔQ_{t-1} , and ΔD_{t-1} .

The numerical values of the lower bound $\underline{p}$ and upper bound $\bar{p}$ are strictly positive at the 1% level for all countries. The 1% level bound $\underline{p}$ is 0.1% for all countries. The 1 percent level bound $\bar{p}$ ranges from a low of 1.7% in Denmark to a high of 8.9% in Norway. At other levels, the bounds are not always defined. The GMM estimates $\hat{p}$ fall inside the 1% bounds for all countries but Finland. Although the birth rate estimates $\hat{p}$ are imprecise, the overidentifying restrictions related to the GMM estimates are never rejected. The implication is that the orthogonality restrictions statistically hold for all countries. Finally, the actual birth rates $\wp$ range from a low of 1.0% per year in Germany to a high of 1.5% per year in Australia, Canada, and the United States. The actual birth rates $\wp$ fall inside the 1% level bounds $\underline{p}$ and $\bar{p}$ for all countries. This means that the restrictions of the model hold for the actual birth rate $\wp$ at the 1% level for all countries.

As discussed above, the orthogonality restrictions are not rejected with the actual birth rate. This suggests that the model adequately fits the data. This conclusion is confirmed by comparing the actual adjusted current account x_t computed directly from the data to the fitted adjusted current account $\hat{x}_t$ computed from the restricted reduced form (18), where $r = 0.04$ and $p = \wp$. Figure 2 highlights that the fit is overall good, given the simplicity of the model. The fit is very good for some countries (e.g., Finland, Germany, Sweden, and Switzerland), but perhaps not as good for a few countries (e.g., Canada and the United Kingdom).

3.3. Tax shocks

Finally, our analysis relies on evaluating the relation between the budget and external deficits generated by tax shocks. The vector error correction specifications

TABLE 2
Birth rates

Country	Level	p	$\bar{p}$	$\hat{p}$	φ
Australia	1	0.001	0.032	0.021	0.015
	5	0.001	0.007	(0.019)	
	10	—	—	[0.630]	
Austria	1	0.001	0.042	0.032	0.012
	5	0.001	0.028	(0.033)	
	10	0.001	0.013	[0.119]	
Canada	1	0.001	0.076	0.025	0.015
	5	0.001	0.055	(0.010)	
	10	0.001	0.039	[0.324]	
Denmark	1	0.001	0.017	0.009	0.012
	5	0.001	0.007	(0.011)	
	10	0.001	0.003	[0.782]	
Finland	1	0.001	0.019	0.029	0.013
	5	0.001	0.003	(0.013)	
	10	—	—	[0.367]	
France	1	0.001	0.074	0.027	0.014
	5	0.001	0.052	(0.031)	
	10	0.001	0.038	[0.838]	
Germany	1	0.001	0.043	0.039	0.010
	5	0.001	0.030	(0.032)	
	10	—	—	[0.353]	
Italy	1	0.001	0.079	0.056	0.011
	5	0.001	0.058	(0.062)	
	10	0.001	0.046	[0.583]	
Japan	1	0.001	0.043	0.008	0.011
	5	0.001	0.009	(0.007)	
	10	—	—	[0.534]	
Netherlands	1	0.001	0.066	0.016	0.013
	5	0.001	0.034	(0.020)	
	10	—	—	[0.413]	
Norway	1	0.001	0.089	0.029	0.013
	5	0.001	0.044	(0.047)	
	10	0.001	0.001	[0.125]	
Spain	1	0.001	0.074	0.032	0.013
	5	—	—	(0.025)	
	10	—	—	[0.655]	
Sweden	1	0.001	0.055	0.006	0.012
	5	0.001	0.012	(0.008)	
	10	—	—	[0.872]	
Switzerland	1	0.001	0.085	0.029	0.011
	5	—	—	(0.018)	
	10	—	—	[0.240]	

(continued)

TABLE 2
(Continued)

Country	Level	$\underline{p}$	$\bar{p}$	$\hat{p}$	$\wp$
U.K.	1	0.001	0.020	0.001	0.013
	5	—	—	(0.014)	
	10	—	—	[0.108]	
U.S.	1	0.001	0.061	0.037	0.015
	5	0.001	0.031	(0.052)	
	10	—	—	[0.576]	

NOTES: Entries are the estimated and actual values of the birth rate. $\underline{p}$ and $\bar{p}$ are the smallest and largest values of the birth rate for which the orthogonality restrictions are not rejected at the 1, 5, and 10% levels of significance. $\hat{p}$ represents the GMM estimates of the birth rate. $\wp$ is the actual annual birth rate. Numbers in parentheses are the standard errors of the GMM estimates. Entries in brackets are the probability values associated with the χ^2 -test of overidentifying restrictions.

can be used for this evaluation because they permit the construction of artificial data series that condition on tax shocks. This requires the identification of orthogonal tax shocks. The challenge here is that the existence of automatic fiscal stabilizers implies that changes in net output mechanically alter net tax revenues. For example, tax revenues automatically decline when an economy moves into a recession because of higher unemployment and lower income, whereas transfers increase, owing to larger state-contingent payments, such as unemployment insurance and aid to dependants. For our analysis, we wish to purge these effects of the business cycle on net taxes to retain only the exogenous portion of unexpected net tax revenues. Fortunately, a simple identification strategy that purges the effects of automatic stabilizers and remains consistent with the structural model consists of applying a Cholesky decomposition. Note that a similar approach is taken in Kim and Roubini (2008). Eventually, the robustness of the results will be verified using the alternative identification scheme proposed by Blanchard and Perotti (2002) (see the technical appendix, A.5.)

For the Cholesky decomposition, consider $\Xi_{x,t} = \Lambda_x^{-1} \mathbf{U}_{x,t}$, where $\Xi_{x,t}$ is the vector of shocks, $\mathbf{U}_{x,t}$ is the vector of statistical innovations, Λ_x is a lower triangular matrix, $E(\mathbf{U}_{x,t} \mathbf{U}_{x,t}') = \Lambda_x \Lambda_x'$, and $E(\Xi_{x,t} \Xi_{x,t}') = \mathbf{I}$. The decomposition imposes the following relations

$$\lambda_{11}^* u_{Q,t} = \xi_{Q,t} \quad (21)$$

$$\lambda_{22}^* u_{D,t} = -\lambda_{21}^* u_{Q,t} + \xi_{T,t} \quad (22)$$

$$\lambda_{33}^* u_{x,t} = -\lambda_{31}^* u_{Q,t} - \lambda_{32}^* u_{D,t} + \xi_{x,t}, \quad (23)$$

where $\mathbf{U}_{x,t} = (u_{Q,t} \ u_{D,t} \ u_{x,t})'$, $\Xi_{x,t} = (\xi_{Q,t} \ \xi_{T,t} \ \xi_{x,t})'$, and λ_{ij}^* is the (i, j) element of Λ_x^{-1} .

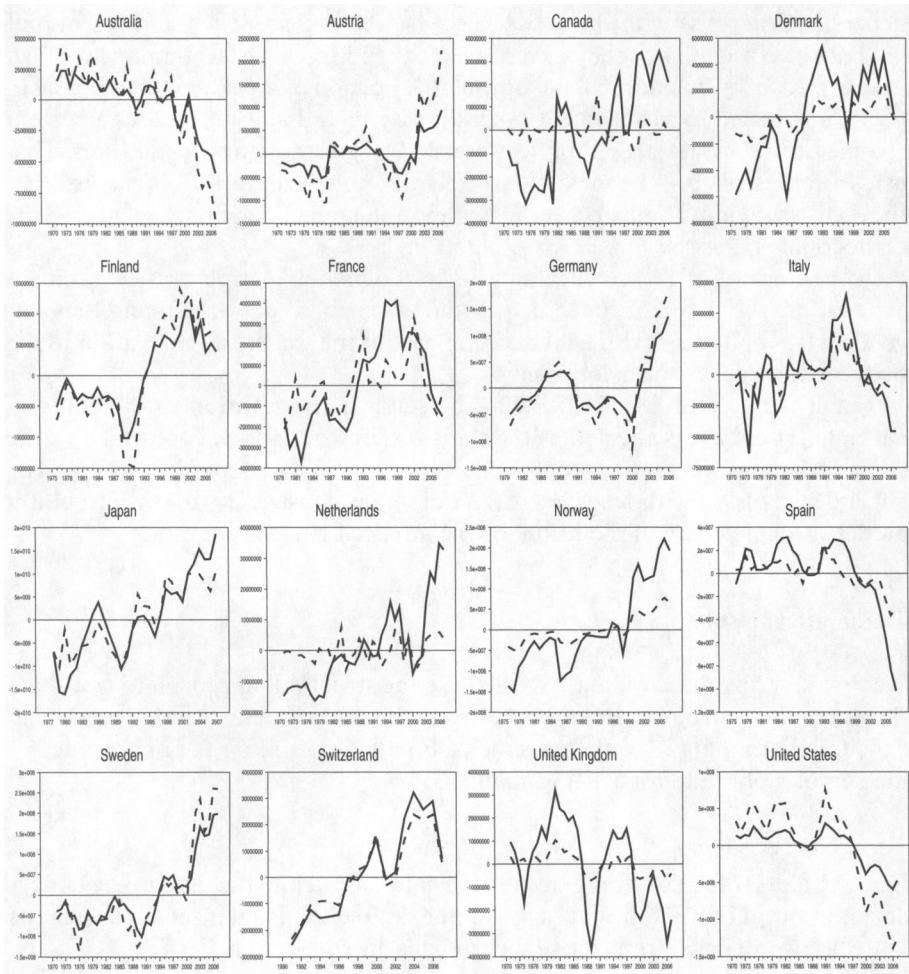


FIGURE 2 Actual and predicted adjusted current account

NOTES: The solid lines show the actual adjusted current account s_t . The dashed lines display the fitted values of the adjusted current account $\hat{s}_t$ generated by the restricted VECMs. All computations used the actual values of the birth rate and the unrestricted law of motion (FULL).

The above decomposition is consistent with the structural model of section 2. For example, equation (21) implies that the statistical innovation to net output $u_{Q,t}$ depends only on the orthogonal shock $\xi_{Q,t}$. This is consistent with the model, because net output is unaffected at impact by tax policies and current account movements (see the technical appendix, A.2).

Equation (22) implies that the statistical innovation to the budget deficit $u_{D,t}$ is affected by the statistical innovation to net output $u_{Q,t}$ and an orthogonal shock $\xi_{T,t}$. The dependence of $u_{D,t}$ with respect to $u_{Q,t}$ reflects changes in the budget deficit that are related to changes in the economic condition. This

dependence accords with the model, because the budget deficit and net output are both affected at impact by government expenditures (i.e., a component of net output). Also, by definition of net output, the portion of $u_{D,t}$ that is orthogonal to $u_{Q,t}$ is unaffected by government expenditures, such that $\xi_{T,t}$ captures a current change of the budget deficit that is unrelated to government expenditures. That is, $\xi_{T,t}$ represents a tax shock. Put differently, the decomposition ensures that the statistical innovations to net output and the budget deficit are related to the orthogonal shocks by $u_{Q,t} = \lambda_{11}\xi_{Q,t}$ and $u_{D,t} = \lambda_{21}\xi_{Q,t} + \lambda_{22}\xi_{T,t}$, where λ_{ij} is the (i, j) element of Λ_x . Accordingly, as in the model, a structural government expenditure shock is embedded in $\xi_{Q,t}$ and affects both net output and the budget deficit simultaneously. The remaining part of the statistical innovation to the budget deficit comes from a tax shock $\xi_{T,t}$.

Finally, equation (23) implies that the statistical innovation to the adjusted current account $u_{x,t}$ is affected by the statistical innovation to net output $u_{Q,t}$, the statistical innovation to the budget deficit $u_{D,t}$, and an orthogonal shock $\xi_{x,t}$. The dependence of $u_{x,t}$ with respect to $u_{Q,t}$ and $u_{D,t}$ is consistent with the structural model, as depicted by the solution for the adjusted current account.

4. Empirical results

We first describe the raw data. We then use the structural approach to construct the contemporaneous correlation between the budget deficit and the external deficit, conditioning on only tax shocks. Finally, we explore the importance of the demographic and forecasting channels.

4.1. Data description

Figure 1 plots the ratio of the external deficit to output and the ratio of the budget deficit to output for all the countries in our sample. A visual inspection suggests that the two deficits are positively correlated in some countries (e.g., Canada, Germany, and Norway), but negatively correlated in other countries (e.g., Spain). Clearly, the figure shows that the experiences are diverse. For example, the budget deficits of Australia and Canada follow similar patterns. The budget deficits are low in the early 1970s, but rise considerably in the early to mid-1980s. The deficits then decline in the late 1980s to rise considerably in the mid-1990s, and finally decline to remain low in the mid-2000s. In contrast, for the same time period, the external deficit of Australia fluctuates, but displays a clear increase in the early period to remain high until the end of the sample. The external deficit of Canada rises in the early period, drops in the mid-1980s, bounces back up in the late 1980s, remains high through the mid-1990s, and then declines to remain low until the end of the sample. These patterns suggest a low correlation between the two deficits for Australia, but a fairly high correlation for Canada.

Table 3 provides GMM estimates of the contemporaneous correlation between the budget deficit ratio and the external deficit ratio from the raw data. We

TABLE 3
Correlations between budget and external deficit ratios

Country	RAW	Unrestricted			Restricted		
		VECM	VAR	BP	FULL	FCST	DEMO
Australia	−0.052 (0.151) [0.732]	0.587 (0.176) [0.001]	0.666 (0.101) [0.000]	0.366 (0.126) [0.004]	0.738 (0.073) [0.000]	0.686 (0.064) [0.000]	0.938 (0.013) [0.000]
Austria	−0.035 (0.059) [0.549]	0.885 (0.035) [0.000]	0.810 (0.059) [0.000]	0.503 (0.070) [0.000]	0.386 (0.178) [0.000]	0.290 (0.086) [0.001]	0.987 (0.003) [0.000]
Canada	0.544 (0.102) [0.000]	0.515 (0.129) [0.000]	0.306 (0.139) [0.028]	0.349 (0.165) [0.034]	0.976 (0.005) [0.000]	0.690 (0.013) [0.000]	0.944 (0.016) [0.000]
Denmark	0.152 (0.182) [0.405]	0.563 (0.136) [0.000]	0.561 (0.165) [0.001]	0.448 (0.185) [0.015]	0.871 (0.044) [0.000]	0.849 (0.059) [0.000]	0.912 (0.021) [0.000]
Finland	−0.139 (0.162) [0.390]	0.418 (0.159) [0.008]	0.401 (0.153) [0.008]	0.180 (0.159) [0.259]	0.901 (0.033) [0.000]	0.166 (0.149) [0.266]	0.957 (0.010) [0.000]
France	−0.144 (0.129) [0.262]	0.772 (0.051) [0.000]	0.513 (0.122) [0.000]	0.441 (0.146) [0.002]	0.719 (0.072) [0.000]	0.702 (0.073) [0.000]	0.885 (0.031) [0.000]
Germany	0.415 (0.180) [0.021]	0.401 (0.249) [0.108]	0.540 (0.191) [0.005]	0.412 (0.199) [0.038]	0.946 (0.022) [0.000]	0.209 (0.183) [0.252]	0.981 (0.006) [0.000]
Italy	0.020 (0.146) [0.889]	0.408 (0.114) [0.000]	0.286 (0.161) [0.076]	0.514 (0.145) [0.000]	0.965 (0.010) [0.000]	0.982 (0.004) [0.000]	0.916 (0.020) [0.000]
Japan	−0.293 (0.106) [0.006]	0.685 (0.093) [0.000]	0.561 (0.104) [0.000]	0.713 (0.080) [0.000]	0.938 (0.016) [0.000]	0.906 (0.004) [0.000]	0.972 (0.008) [0.000]
Netherlands	−0.017 (0.114) [0.883]	0.459 (0.131) [0.001]	0.782 (0.058) [0.000]	0.804 (0.054) [0.000]	0.973 (0.011) [0.000]	0.666 (0.099) [0.000]	0.964 (0.011) [0.000]
Norway	0.648 (0.092) [0.000]	0.774 (0.071) [0.000]	0.088 (0.286) [0.757]	0.633 (0.112) [0.000]	0.739 (0.113) [0.000]	0.621 (0.118) [0.000]	0.973 (0.008) [0.000]
Spain	−0.818 (0.058) [0.000]	0.463 (0.133) [0.001]	0.212 (0.166) [0.203]	0.345 (0.145) [0.017]	0.917 (0.022) [0.000]	0.898 (0.027) [0.000]	0.921 (0.026) [0.000]
Sweden	0.137 (0.121) [0.255]	0.904 (0.032) [0.000]	0.783 (0.048) [0.000]	0.831 (0.035) [0.000]	0.995 (0.001) [0.000]	0.997 (0.001) [0.000]	0.916 (0.022) [0.000]
Switzerland	0.210 (0.211) [0.319]	0.602 (0.156) [0.000]	0.417 (0.186) [0.025]	0.455 (0.152) [0.003]	0.717 (0.092) [0.000]	0.243 (0.184) [0.187]	0.991 (0.005) [0.000]

(continued)

TABLE 3
(Continued)

Country	RAW	Unrestricted			Restricted		
		VECM	VAR	BP	FULL	FCST	DEMO
U.K.	0.083 (0.178) [0.641]	0.899 (0.030) [0.000]	0.912 (0.023) [0.000]	0.771 (0.095) [0.000]	0.524 (0.096) [0.000]	0.286 (0.138) [0.038]	0.972 (0.009) [0.000]
U.S.	-0.057 (0.144) [0.694]	0.589 (0.098) [0.000]	0.321 (0.152) [0.034]	0.256 (0.152) [0.092]	0.949 (0.013) [0.000]	0.511 (0.111) [0.000]	0.988 (0.003) [0.000]

NOTE: Entries are the GMM estimates of the contemporaneous correlation between the ratios of the budget and external deficits to output. Raw refers to the estimates computed from the raw data. Unrestricted refers to the estimates computed using unrestricted VECMs conditioning on tax shocks. For the unrestricted VECMs, the shocks are identified with a Cholesky decomposition (VECM). For the VECMs that omit the error correction term, the shocks are identified either via a Cholesky decomposition (VAR) or the Blanchard and Perotti approach (BP). Restricted refers to the estimates computed from the restricted VECMs with a Cholesky decomposition to identify tax shocks. The computations use the actual birth rate and the unrestricted law of motion (FULL), a birth rate of 0.001 and the unrestricted law of motion (FCST), and the actual birth rate and univariate AR laws of motion (DEMO). Numbers in parentheses are the standard errors of the estimates and entries in brackets are the probability values associated with the test that the correlation is null.

compute these correlations for the ratios of deficits to output rather than for the levels of deficits, because the levels of deficits are integrated processes. The two deficits are positively correlated in some countries and negatively correlated in others. The unconditional correlation ranges from a low of -0.818 for Spain to a high of 0.648 in Norway. It is negative for eight countries and positive for eight countries. As expected, the correlation between the two deficits is near zero for Australia (-0.052) and high for Canada (0.544).

So far, the data show little evidence of a general pattern of twin deficits. Importantly, the raw data correlations jointly capture the effects of several shocks, and do not provide enough information to isolate the effects of specific shocks such as tax cuts.

4.2. Tax cuts and twin deficits: unrestricted VECMs

To disentangle the effects of tax shocks, we compute the GMM estimates of the contemporaneous correlation between the budget deficit ratio and the external deficit ratio using simulated series constructed from estimated versions of the unrestricted laws of motion (17) and (19). In doing so, we use a Cholesky decomposition to retain only tax shocks (see the technical appendix, A.6). For this analysis, we assume that the world interest rate is $r = 0.04$ and that the birth rate is the actual birth rate $p = \wp$ for each country.

Table 3 reports the estimates of the conditional contemporaneous correlation. The correlations for this first experiment appear under the heading VECM.

The estimates are positive, numerically large, and statistically significant. The estimates range from a low of 0.401 for Germany to a high of 0.904 for Sweden. The estimates are statistically significant at the 10% levels for all countries, except Germany.

To verify the robustness of these results, table 3 also presents the estimates of the contemporaneous correlation computed for two different exercises. The first exercise eliminates the error correction term to the unrestricted laws of motion (17) and (19), but preserves the Cholesky decomposition to identify tax shocks. The correlations for this experiment appear under VAR. The estimates range from a low of 0.088 for Norway to a high of 0.912 for the United Kingdom. Note that the estimates are above 0.4 for 11 out of the 16 countries. Also, the estimates are statistically significant at the 10% level for all countries but Norway and Spain.

The second exercise also omits the error correction term, and uses a different strategy to identify tax shocks. For this, we employ a variant of the Blanchard and Perotti (2002) strategy, which adds the adjusted current account to the analysis. A similar approach is taken in Monacelli and Perotti (2010). The results of this experiment appear under the heading BP. The estimates of the contemporaneous correlation obtained with the alternative decomposition are also positive, large, and statistically significant. The estimates range from a low of 0.180 for Finland to a high of 0.831 for Sweden. Also, the estimates are above 0.4 for 11 out of the 16 countries and statistically significant at the 10% level for all countries, but Finland.

Overall, the estimates of the correlation between the budget deficit ratio and the external deficit ratio obtained from the unrestricted laws of motion are mostly positive, large, and significant. This is so whether or not we include the error correction term and whether we use the Cholesky decomposition or the Blanchard-Perotti identification strategy. These correlations display a general pattern of twin deficits. Unfortunately, these estimates ignore some of the restrictions of the structural model and thus cannot be used to study the different channels by which tax shocks affect both deficits.

4.3. Tax cuts and twin deficits: restricted VECMs

To fully accord with the structural model, the GMM estimates of the contemporaneous correlation between the budget deficit ratio and the external deficit ratio are also computed from the restricted reduced forms (18) and (20) with the Cholesky decomposition (see the technical appendix, A.6).

Table 3 reports the estimates of the conditional correlations. The results for this experiment based on $r = 0.04$ and $p = \varnothing$ appear under the heading FULL. The positive correlations are numerically large and significantly different from zero. The correlations range from a low of 0.386 for Austria to a high of 0.995 for Sweden. The estimates are statistically significant at any conventional level.

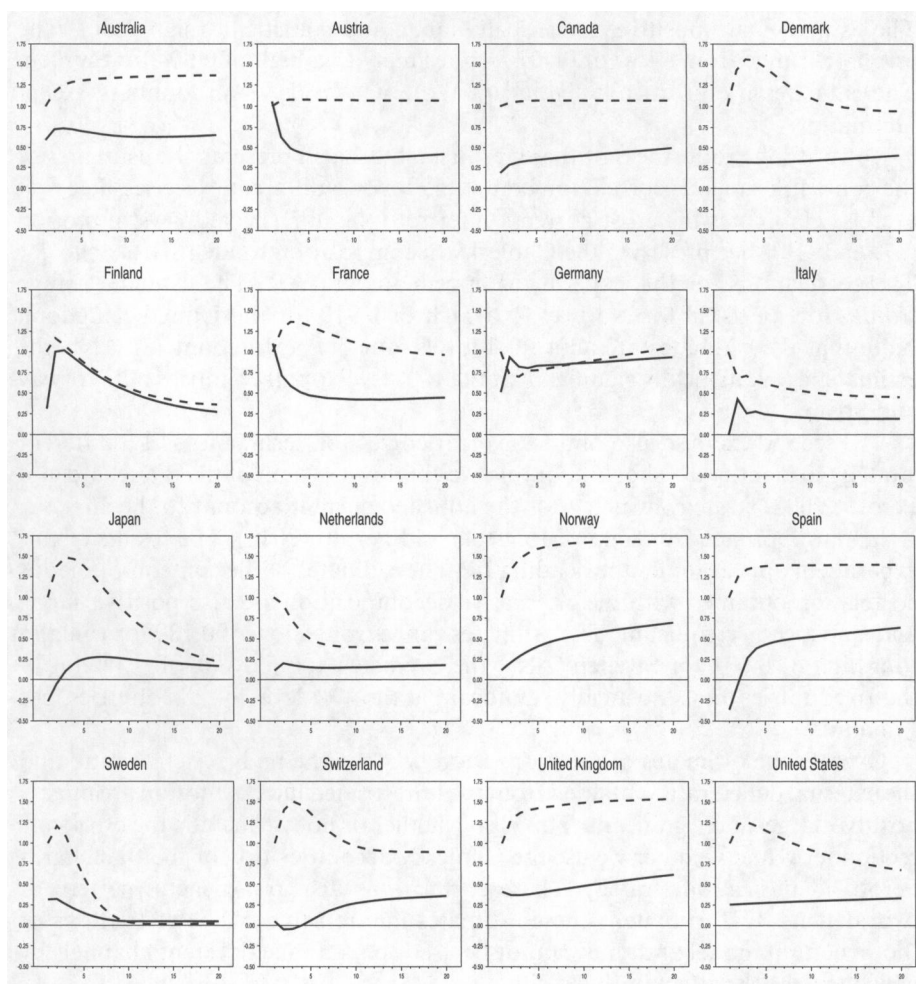


FIGURE 3 Dynamic responses of deficit levels: FULL

NOTES: The solid lines correspond to the dynamic responses of the external deficit level and the dashed lines correspond to the dynamic responses of the budget deficit level. All computations are based on the actual values of the birth rates and the unrestricted law of motion (FULL).

Thus far, the country estimates of the correlation between the budget and external deficits constructed from the restricted reduced forms support a general pattern of twin deficits. Before we study the importance of the different channels by which tax shocks affect the external deficit, it is worthwhile to understand the general mechanisms at play. Intuitively, the positive correlation between the budget and external deficits can be attributed to the positive responses of both the budget and external deficits to a tax cut over most horizons. Accordingly, figure 3 displays the dynamic responses of the levels of the budget and external

deficits to a tax cut (see the technical appendix, A.6). That is, the figure shows the dynamic responses to an unexpected currency unit increase of the budget deficit, which is exclusively due to an unexpected tax cut, constructed from the restricted reduced forms (18) and (20). Also, we note that the responses are for the level of the budget and external deficits, not for ratios of the budget and external deficits to net output (that were used to compute the conditional correlations).

By design, the tax cut generates a unit positive impact response of the budget deficit for all countries. The positive response of the budget deficit is highly persistent for all countries but the Netherlands and Sweden. The tax cut also generates a large positive impact response of the external deficit for most countries. Exceptions include the negative impact response for Japan and Spain and the near zero response for Italy and Switzerland. For Italy, Japan, and Spain, the external deficit rises sharply soon after impact. For Switzerland, the responses eventually rise after a few years. We also note the very large positive impact response for Austria and France, where the response of the external deficit is slightly more than unity. Finally, the response of the external deficit appears very persistent for many countries, but not Sweden.

It is also worthwhile to verify whether the tax shock is an important contributor to variations in the budget and external deficits. To this end, table 4 presents a forecast error variance decomposition for the level of the budget and external deficits (not for the ratios to net output). Specifically, entries are the percentage proportions of the forecast error variances at different horizons (1, 5, and 10 years) that are attributable to the tax shock.

The results of this experiment appear under FULL. As is expected, the tax shock accounts for a very large fraction of the variance of the forecast error for the level of the budget deficit at a one year horizon, but explains less of the variance as the horizon extends. This simply results from the estimated dynamics of the restricted reduced forms. More interestingly, the tax shock explains a sizeable fraction of the variance of the forecast error for the external deficit. This is true for most countries but Austria, Germany, Norway, and Switzerland. After a decade, the fraction ranges from a low of 2.808% for Germany to a high of 88.246% for France. As an example, consider the case of Canada. The tax shock accounts for a large fraction of the forecast error variance at all horizons. It accounts for 49.25% at a 1 year horizon, 51.24% at the 5 year horizon, and 47.66% at the 10-year horizon.

Finally, it is useful to assess whether the tax cut generates economically substantial budget and external deficits. Accordingly, table 5 presents the effects of a currency unit tax cut on the levels of the public and external debt, where the debt responses are obtained by cumulating the responses of the deficits. These are the cumulated responses of the levels of deficits after a year, after five years, and after a decade. The results appear under the heading FULL.

After a year, the tax cut must generate a unit rise in the level of public debt. After a decade, the tax cut generates a substantial increase in the level of the public debt for all countries. The responses range from a low of about 5.6 currency units

TABLE 4
Variance decomposition for budget and external deficit levels

FULL				FCST		DEMO	
Country	Years	Budget	External	Budget	External	Budget	External
Australia	1	99.308	6.034	98.939	10.074	99.545	46.528
	5	79.647	10.614	78.319	15.218	99.545	81.981
	10	73.355	13.997	74.239	15.823	99.545	91.359
Austria	1	90.212	2.962	92.167	0.013	94.947	2.185
	5	82.407	2.372	88.149	0.468	94.947	10.666
	10	78.967	2.924	87.556	0.450	94.947	19.704
Canada	1	68.614	49.247	68.489	5.908	72.874	71.842
	5	50.295	51.242	52.315	5.616	72.874	76.086
	10	47.880	47.658	52.897	7.219	72.874	73.492
Denmark	1	88.706	51.414	90.207	0.524	86.691	73.331
	5	81.329	53.757	83.804	3.217	86.691	84.402
	10	65.114	47.870	74.056	14.440	86.691	85.770
Finland	1	82.632	2.558	80.728	0.478	83.067	65.504
	5	74.216	24.546	73.489	7.804	83.067	81.257
	10	56.208	30.876	58.273	10.494	83.067	81.791
France	1	99.734	88.675	99.606	22.959	98.857	96.352
	5	93.943	89.779	83.243	10.838	98.857	98.486
	10	80.513	88.246	53.276	8.291	98.857	98.704
Germany	1	87.376	1.156	83.955	13.278	87.567	88.719
	5	46.070	2.665	31.443	13.009	87.567	87.753
	10	32.167	2.808	16.696	11.977	87.567	87.640
Italy	1	96.229	0.006	96.476	23.823	97.995	53.522
	5	46.597	17.981	63.412	18.388	97.995	84.872
	10	20.839	24.365	54.646	18.776	97.995	91.937
Japan	1	99.825	22.374	99.994	9.878	99.952	44.850
	5	81.973	26.892	75.001	3.659	99.952	80.947
	10	50.775	28.093	43.919	3.541	99.952	90.972
Netherlands	1	99.979	22.635	99.993	0.604	99.997	74.190
	5	46.148	27.546	43.751	29.978	99.997	94.253
	10	24.401	17.878	21.741	19.968	99.997	97.487
Norway	1	30.032	1.350	30.006	3.323	30.169	95.605
	5	38.514	5.976	36.900	1.167	30.169	35.487
	10	50.436	16.987	48.124	1.088	30.169	31.928
Spain	1	98.544	75.261	98.537	63.052	99.396	65.596
	5	95.775	80.129	96.029	44.488	99.396	91.353
	10	95.219	86.082	95.708	33.160	99.396	95.780
Sweden	1	71.297	30.936	67.543	3.899	73.814	74.082
	5	35.202	24.941	50.916	23.521	73.814	73.858
	10	16.009	12.583	41.617	30.450	73.814	73.831
Switzerland	1	97.785	0.301	99.554	4.683	95.832	99.915
	5	85.734	0.457	84.702	16.764	95.832	96.570
	10	63.522	7.337	63.433	17.079	95.832	96.094

(continued)

TABLE 4
(Continued)

FULL				FCST		DEMO	
Country	Years	Budget	External	Budget	External	Budget	External
U.K.	1	96.876	85.318	97.299	96.082	97.963	72.334
	5	79.482	83.632	84.025	80.977	97.963	91.813
	10	57.420	61.853	65.268	64.674	97.963	95.379
U.S.	1	91.914	54.724	91.639	5.865	94.952	68.059
	5	64.459	47.629	65.547	13.551	94.952	90.366
	10	56.892	42.226	60.915	21.098	94.952	92.626

NOTES: Entries are the percentage contributions of the tax shock to the forecast error variances of the levels of the budget and external deficits. The computations are based on versions with the actual birth rate and the unrestricted law of motion (FULL), with a birth rate of 0.001 and the unrestricted law of motion (FCST), and with the actual birth rate and univariate AR laws of motion (DEMO).

for Sweden to a high of 13.8 currency units for the United Kingdom. Overall, a tax cut generates a more than tenfold increase in the level of the public debt for 11 countries. As an example, a dollar increase in the level of the budget deficit of Canada generates an \$11 public debt after 10 years.

After a year, the tax cut generates a rise in the level of the external debt for 13 countries. After a decade, the responses of the external debt are positive for all countries. The responses of the external debt after a decade range from a low of about 1 currency unit for Switzerland to a high of more than 7 currency units for Finland. Note that the responses of the external debt are lower than those of the public debt are all countries but Germany. To pursue our previous example, consider the case of Canada. After a decade, a \$1 tax cut generates a level of external debt of roughly \$3.

Overall, these results provide evidence that a tax cut generates twin deficits. The positive conditional correlations between the budget and external deficits obtained from the restricted reduced forms are large and significant. The positive dynamic responses of the budget and external deficits describe the mechanism by which the tax cut is propagated through time. Also, the variance decompositions and the dynamic responses of the debts confirm the economic importance of twin deficits. However, this analysis does not provide an evaluation of the different channels by which a tax cut affects the external deficit.

4.4. Demographic versus forecasting channels

To disentangle the effects of the different channels on the propagation of the tax shock, we redo the empirical analysis while shutting down the different channels. In what follows, the correlations and dynamic responses are computed using the restricted reduced forms (18) and (20). To do so, we re-estimate the structural model for the different parameterizations. The re-estimation implies

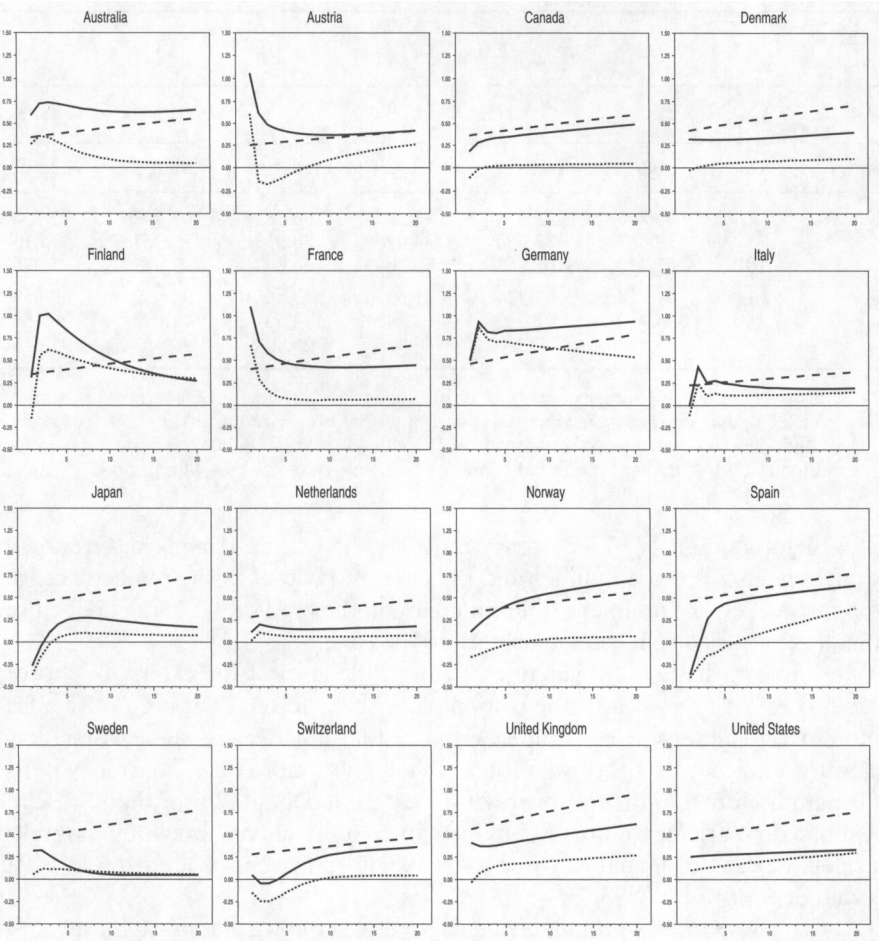


FIGURE 4 Dynamic responses of deficit levels: DEMO vs. FCST
NOTES: The solid lines correspond to the dynamic responses of external deficit level for the version with the actual birth rate and the unrestricted law of motion (FULL). The dotted lines show the dynamic responses of external deficit level for the version with a birth rate of 0.001 and the unrestricted law of motion (FCST). The dashed lines represent the dynamic responses of external deficit level for the version with the actual birth rate and univariate AR laws of motion (DEMO).

small differences in the responses of the budget deficit to the tax shock, but the responses remain positive at all horizons.

For tables 3, 4, and 5, as well as figure 4, the results under forecasting (FCST) are computed for a version of the full structural model that closes the demographic channel. For this, the birth rate is set to $p = 0.001$ for all countries. Although this value is not strictly null, it is small enough to ensure that the demographic channel is negligible. With this parameterization, consumers find it very difficult to shift their tax burden to future generations.

TABLE 5
Cumulative responses of budget and external deficit levels

FULL				FCST		DEMO	
Country	Years	Budget	External	Budget	External	Budget	External
Australia	1	1.000	0.595	1.000	0.259	1.000	0.342
	5	6.006	3.494	4.655	1.249	5.828	1.838
	10	12.639	6.793	10.631	2.095	11.922	3.986
Austria	1	1.000	1.058	1.000	0.603	1.000	0.252
	5	4.226	3.033	4.099	0.140	5.202	1.353
	10	10.669	4.925	9.309	0.159	10.457	2.934
Canada	1	1.000	0.186	1.000	-0.109	1.000	0.365
	5	5.231	1.465	5.366	-0.069	6.139	1.965
	10	10.568	3.355	10.895	0.097	10.682	4.263
Denmark	1	1.000	0.322	1.000	-0.025	1.000	0.421
	5	5.530	1.593	8.671	0.096	7.654	2.270
	10	12.080	3.231	15.532	0.432	16.763	4.942
Finland	1	1.000	0.311	1.000	-0.149	1.000	0.345
	5	5.120	4.109	5.745	2.161	6.368	1.856
	10	8.496	7.290	11.037	4.468	13.256	4.030
France	1	1.000	1.102	1.000	0.669	1.000	0.404
	5	4.997	3.372	7.085	1.302	6.874	2.171
	10	11.152	5.548	13.611	1.622	14.571	4.716
Germany	1	1.000	0.512	1.000	0.397	1.000	0.461
	5	3.182	3.066	3.232	2.807	6.795	2.487
	10	7.253	7.279	7.479	6.213	10.405	5.426
Italy	1	1.000	-0.007	1.000	-0.125	1.000	0.225
	5	3.649	1.199	3.266	0.370	5.018	1.211
	10	6.371	2.274	6.938	0.948	10.041	2.627
Japan	1	1.000	-0.265	1.000	-0.366	1.000	0.432
	5	6.702	0.245	7.075	-0.424	8.112	2.329
	10	11.787	1.576	13.289	0.066	18.157	5.076
Netherlands	1	1.000	0.111	1.000	0.017	1.000	0.287
	5	3.723	0.799	3.902	0.366	5.506	1.541
	10	5.798	1.537	6.238	0.684	11.160	3.342
Norway	1	1.000	0.120	1.000	-0.165	1.000	0.339
	5	5.333	1.372	5.205	-0.475	6.283	1.183
	10	13.493	3.884	12.978	-0.390	13.040	3.958
Spain	1	1.000	-0.360	1.000	-0.395	1.000	0.461
	5	6.341	0.257	6.572	-0.909	7.809	2.488
	10	13.289	3.166	13.175	-0.625	17.226	5.415
Sweden	1	1.000	0.316	1.000	0.055	1.000	0.458
	5	4.703	1.245	6.565	0.500	8.069	2.469
	10	5.569	1.615	11.555	0.938	18.023	5.377
Switzerland	1	1.000	0.048	1.000	-0.148	1.000	0.280
	5	6.720	0.019	6.686	-0.984	5.993	1.509
	10	12.127	1.054	12.199	-1.123	12.323	3.277

(continued)

TABLE 5
(Continued)

FULL				FCST		DEMO	
Country	Years	Budget	External	Budget	External	Budget	External
U.K.	1	1.000	0.410	1.000	-0.055	1.000	0.565
	5	6.469	1.932	5.346	0.304	8.809	3.048
	10	13.791	4.176	13.377	1.210	20.455	6.647
U.S.	1	1.000	0.255	1.000	0.100	1.000	0.454
	5	5.847	1.334	6.192	0.644	7.208	2.448
	10	10.865	2.780	12.461	1.586	15.485	5.319

NOTES: Entries are the cumulated responses of the levels of the budget and external deficits to a tax cut. The computations are based on versions with the actual birth rate and the unrestricted law of motion (FULL), with a birth rate of 0.001 and the unrestricted law of motion (FCST), and with the actual birth rate and univariate AR laws of motion (DEMO).

The results under demographic (DEMO) are computed for a version of the structural model that closes the forecasting channel. For this, the law of motion for forcing variables assumes that changes in the budget deficit and in net output follow univariate autoregressive (AR) processes. As a result, the matrices Γ_x and Ψ_w are lower triangular and the error correction terms are absent. This ensures that the current level of the budget deficit contains no information useful to forecast future net output, and this closes the forecasting channel.

The GMM estimates of the conditional correlation between the budget and external deficits suggest that both channels are important for the determination of twin deficits. Recall that, for the forecasting channel, the model does not have a strong prediction concerning the sign of the correlation (see table 1). It is interesting to note that all the estimated correlations are positive. They are also statistically significant for all countries but Finland, Germany, and Switzerland. The correlations range from 0.166 for Finland to 0.997 for Sweden. For the demographic channel, the model predicts non-negative correlations. As expected, the conditional correlations are positive, numerically large, and statistically significant. The conditional correlations range from 0.885 for France to 0.991 for Switzerland.

To help in understanding the transmission of tax shocks, figure 4 shows the dynamic responses of the level of the external deficit obtained under each channel, as well as those obtained with the full structural model. A comparison of these responses suggests that the amplitude of the full model's responses is attributable to the demographic channel, while the shape of the responses is attributable to the forecasting channel. For the forecasting channel, the dynamic responses of the external deficit have similar shapes but smaller amplitudes than the responses generated by the full structural model. The forecasting channel produces a

negative impact response of the external deficit for Canada, Denmark, Finland, Italy, Japan, Norway, Spain, Switzerland, and the United Kingdom. In the cases of Japan and Spain, the negative impact response of the external deficit generated by the forecasting channel is large enough to generate a negative impact response in the full model. For the demographic channel, the dynamic responses of the external deficit are always positive, large, and persistent, but do not display the shapes generated by the full structural model.

Table 4 presents the results from the variance decomposition obtained under each channel. As expected, for both channels, the tax shock explains a large fraction of the forecast error variance for the budget deficit. We note that, by construction, the proportion is constant at different horizon in the demographic case. The tax shock also explains a large fraction of the forecast error variance for the external deficit at all horizons. Admittedly, the fraction explained is much less under the forecasting channel than under the demographic channel.

Likewise, table 5 shows that the impact of the tax cut on the external debt is fairly small and sometimes negative for the forecasting channel, while it is generally large and positive for the demographic channel. In particular, for the forecasting channel, the tax cut generates a reduction in the external debt after a decade for Norway, Spain, and Sweden.

Overall, these results suggest that the demographic and forecasting channels are important in understanding how a tax cut generates twin deficits. Both channels work to produce positive correlations between the budget and external deficits. In particular, the forecasting channel explains the shape of the responses of the external deficit, while the demographic channel explains the amplitude of the responses.

5. Discussion

We study the effects of tax shocks on the budget and external deficits of industrialized economies over the post-1970 period. To do so, we employ a tractable small open economy model with overlapping generations. In the model, a tax cut raises the budget deficit. In turn, the increase in the budget deficit may affect the external deficit via two distinct channels. The demographic channel is standard and it occurs when consumers can shift a portion of their future tax burden to unborn generations. The forecasting channel is less standard, and it occurs when agents require forecasts of future variables to make optimal choices.

We find that tax shocks generate positive comovements between the budget and external deficits for all the countries in our sample. Conditioning on tax shocks, the correlations between the budget and external deficits are large, positive, and significant. These conditional correlations are attributable to the positive responses of both deficits to a tax cut over most horizons. Finally, we find that both demographic and forecasting channels play a role in producing

positive comovements. That is, the forecasting channel explains the shape of the responses of the external deficit, while the demographic channel explains the amplitude of the responses.

Our structural approach relies on a simple model that involves a number of restrictive assumptions. First, it assumes that net output is unrelated to the consumers's saving decisions. Note that this dichotomy is preserved for some extensions of the model. For example, introducing investment adjustment costs alters the dynamics of investment, but investment and net output remain unrelated to saving as long as the interest rate is exogenous. As another example, introducing elastic labour supply via GHH preferences (Greenwood, Hercowitz, and Huffman 1988) alters the dynamics of saving, but net output remains unaffected because GHH preferences suppress the wealth effect on labour supply.

However, the dichotomy between net output and consumers' saving decisions is not preserved when the interest rate is endogenous and when there are wealth effects on labour supply. For example, considering a debt-elastic interest rate in a small open economy model (Schmitt-Grohe and Uribe 2003) or a multi-country general equilibrium model (Boileau and Normandin 2008) implies that net output and saving both depend on the endogenous interest rate.

Second, the model assumes lump-sum taxes. This assumption is obviously not preserved when taxes are distortionary. Finally, our analysis postulates that the 16 industrialized economies in our sample are fairly homogeneous and can be modelled adequately as small open economies. However this assumption may be less appropriate for some of the larger countries in our sample (e.g., the United States).

It would be interesting to refine our analysis by relaxing some of these assumptions. In particular, it would appear important to consider models where the dichotomy between net output and saving is not preserved, where taxes are distortionary, and where countries differ in size.

Data appendix

The annual measures are constructed for the post-1970 period, using raw data from the International Financial Statistics (IFS) released by the International Monetary Fund, the Economic Outlook (EO) published by the Organization for Economic Cooperation and Development, the World Development Indicators (WDI) collected by the World Bank, and Datastream reported by Thomson Reuters. In our analysis, we measure variables in local currencies. For the European countries that have joined the monetary union, the local currency corresponds to the euro. To do so for the pre-1999 period, the national currencies first are converted in US dollars by using the actual nominal exchange rate (source: IFS), and, second, the US dollars are converted in euros by using the synthetic nominal exchange rate (source: Datastream).

The countries (samples) are Australia (1970 to 2007), Austria (1970 to 2007), Canada (1970 to 2007), Denmark (1975 to 2007), Finland (1975 to 2007), France (1978 to 2007), Germany (1979 to 2006), Italy (1970 to 2007), Japan (1977 to 2007), Netherlands (1970 to 2007), Norway (1975 to 2007), Spain (1975 to 2007), Sweden (1970 to 2007), Switzerland (1990 to 2007), the United Kingdom (1970 to 2007), and the United States (1970 to 2007). Germany refers to West Germany and Unified Germany for the pre- and post-1990 periods. Finally, the sample periods are dictated by the availability of the data.

A.1. Current account

The current account is the product of the nominal current account in US dollars (source: IFS) and the actual or synthetic nominal exchange rate of local currency units per US dollar (source: IFS or Datastream), divided by a price index. For Germany the price index is the Gross Domestic Product (GDP) deflator (source: IFS). For the other countries, the price index is the all-item consumer price index (CPI) (source: IFS).

A.2. Budget deficit

The budget deficit is defined as the negative of the nominal budget surplus in local currency divided by the price index. For Germany, the nominal budget surplus is obtained for the Central Government (source: Afonso and Sousa 2009). For the other countries, the nominal budget surplus is collected for the General Government (source: EO).

A.3. Net output

Net output is measured as nominal GDP in local currency (source: IFS) minus the sum of nominal government expenditures in local currency (source: IFS) and nominal investment expenditures in local currency (source: IFS), normalized by the price index.

A.4. Wealth

Wealth is the sum of the nominal net foreign assets in US dollars (source: Lane and Milesi-Ferretti 2007) adjusted by the actual or synthetic nominal exchange rate of local currency units per US dollar (source: IFS or Datastream) and the nominal domestic government debt in local currency, deflated by the price index.

The nominal government debt is measured as the product of the nominal debt service (source: EO) and the reciprocal of the annuity factor $(1 + r)/r$, where the interest rate r is fixed at 4% per year. For Germany, the nominal debt service corresponds to the interest payments on debt of the Central Government (source: Afonso and Sousa 2009). For the other countries, the nominal debt service is the difference between the interest payments and interest receipts of the General Government (source: EO).

A.5. Birth rates

Birth rate refers to the sample average of birth rates (source: WDI).

A.6. Elasticities

The elasticities used to compute the BP decomposition are taken from Girouard and Andre (2005). For each country, the elasticity represents the mean elasticities of corporate tax, personal tax, indirect tax, and social security contributions, where the weights are set to the sample average of the ratio of each type of tax to total tax (source: EO). For Germany and Switzerland, the data start in 1991 and 1990.

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